

Transcriptome Analysis of Diseased Intervertebral Disc Tissue in Patients with Spinal Tuberculosis

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Abstract

Spinal tuberculosis (TB), which is also known as Pott's disease is considered one of the most destructive diseases that affect the spine along with its severe neurological outcomes. In this paper, we investigated transcriptomic alterations in diseased intervertebral disc tissues using RNA-seq data from GEO database under accession GSE268366. Gene differential expression analysis, GSEA analysis, KEGG pathways, and protein-protein interactions were performed to discover molecular pathways and mechanisms involved in the occurrence of spinal TB. The results have shown high enrichment of immune and inflammatory pathways, particularly cytokine signaling and Toll-like receptor signaling. Hub genes and proteins associated with the immune were also identified. These findings provide insights into the molecular basis of spinal tuberculosis it also suggests potential biomarkers and therapeutic targets

Introduction

Tuberculosis continues to be a significant global health issue, resulting from infection by *Mycobacterium tuberculosis*. Spinal tuberculosis can be considered among the most hazardous forms of extrapulmonary tuberculosis due to the involvement of spinal column degeneration, instability, and neurological issues. Molecular mechanisms that are involved in spinal tuberculosis is necessary for developing effective diagnostic and therapeutic ways.

Current advances in RNA sequencing allow for the comprehensive examination of molecular signatures associated with disease development. In this work, RNA transcriptome profiling was performed using the open-access GEO database GSE268366.

Methodology

RNA-seq data were collected from the GEO database under accession number GSE268366. Then data preprocessing and analysis were performed using R.

The Quality assessment included box plots, PCA analysis, and density plots as shown in figures 1-3.

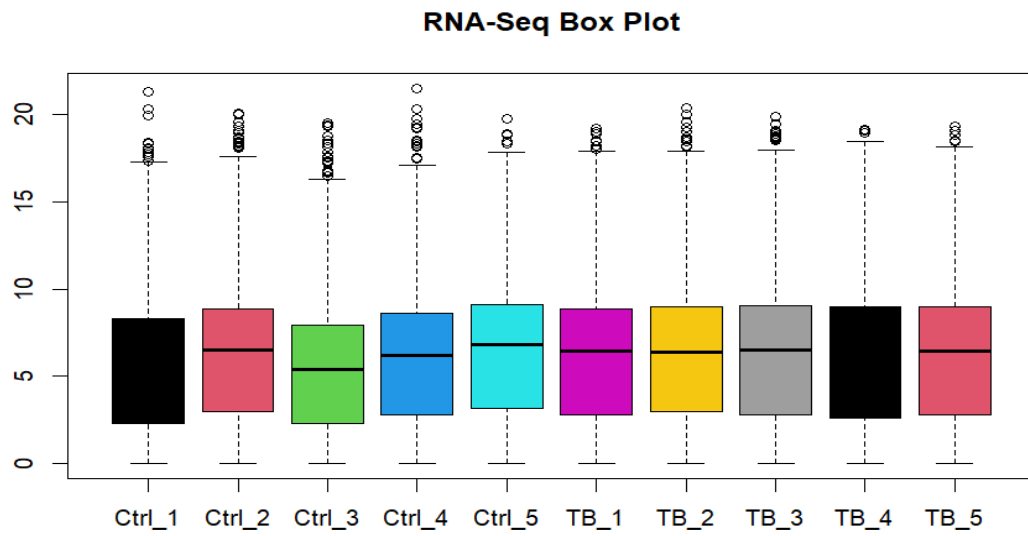


Figure 1. Box plot showing distribution of normalized gene expression values in control and spinal tuberculosis samples.

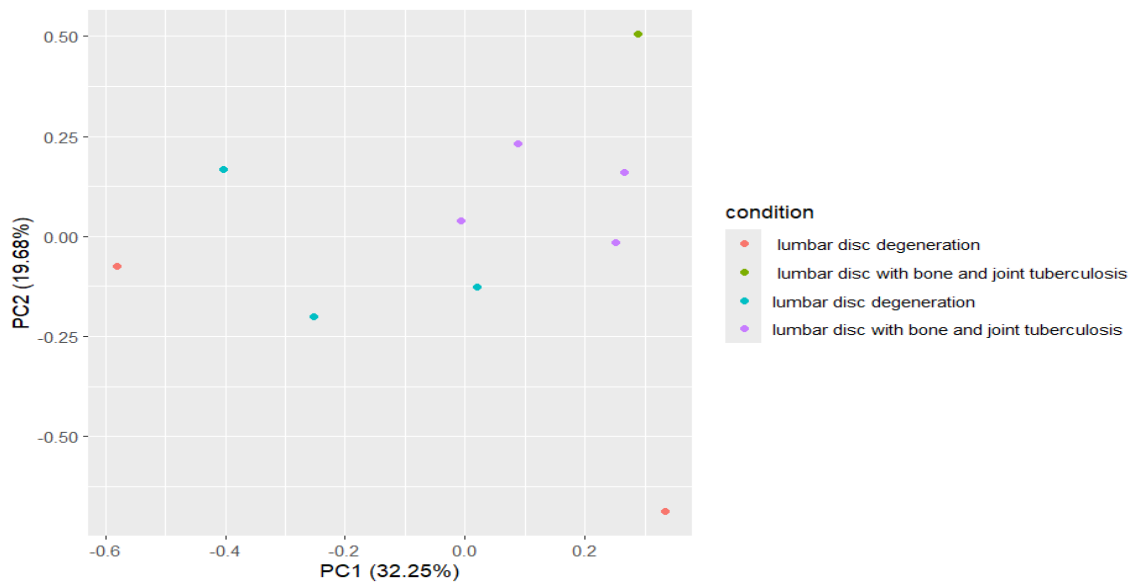


Figure 2. Principal component analysis (PCA) plot for clustering of samples and distribution of variance between spinal tuberculosis and control groups.

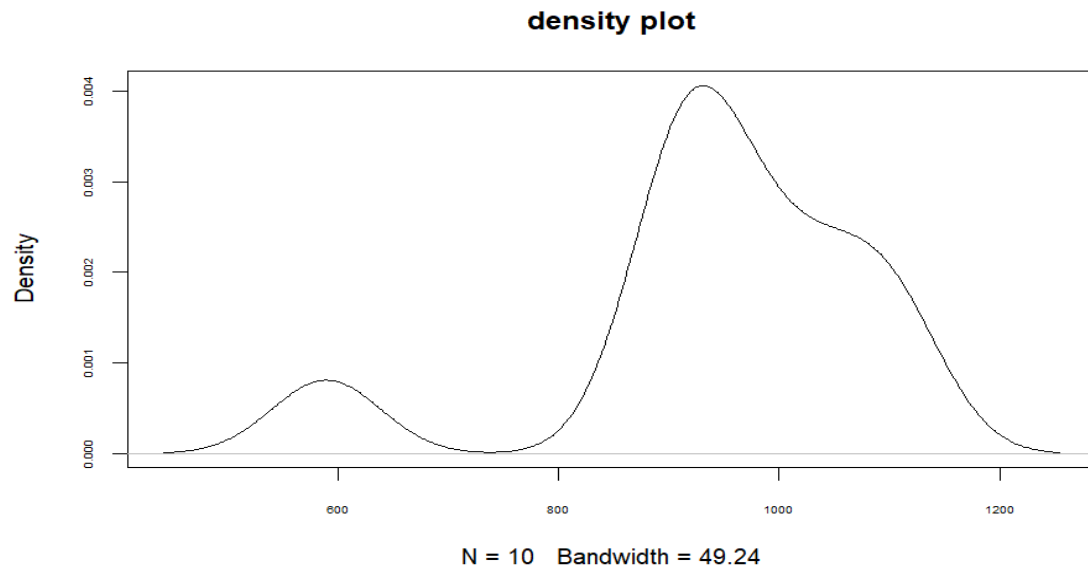


Figure 3. Density plot shows the distribution of normalized expression values across all samples.

The box plot indicated relatively similar expression distributions between samples, indicating acceptable normalization. PCA analysis showed moderate separation between the samples and density plots showed variability among the samples. Differential expression analysis was performed using commonly applied transcriptomic workflows like DESeq2 and visualization packages such as ggplot2. Genes with adjusted p-values < 0.05 and $|\log_2 \text{fold change}| > 1$ were identified as significantly differentially expressed. Functional enrichment analysis included GSEA and KEGG pathway analysis. Protein-protein interaction (PPI) network was performed to identify the hub proteins relevant to spinal tuberculosis.

Results and Discussion

Differential gene expression analysis revealed several differentially expressed genes between spinal TB and lumbar disc degeneration samples. The identified DEGs were mainly associated with immune and inflammatory responses.

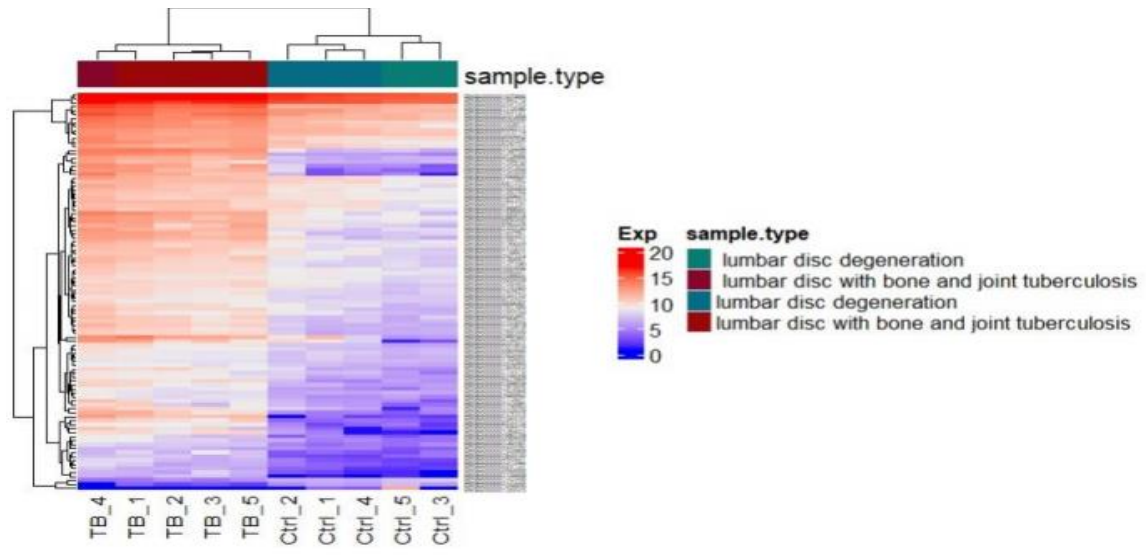


Figure 4. Heatmap of the top 50 differentially expressed genes (DEGs) between spinal tuberculosis and lumbar disc degeneration.

Heatmap analysis showed different expression patterns between spinal TB and control samples. The Volcano plot analysis confirmed transcriptional differences between disease and control groups.

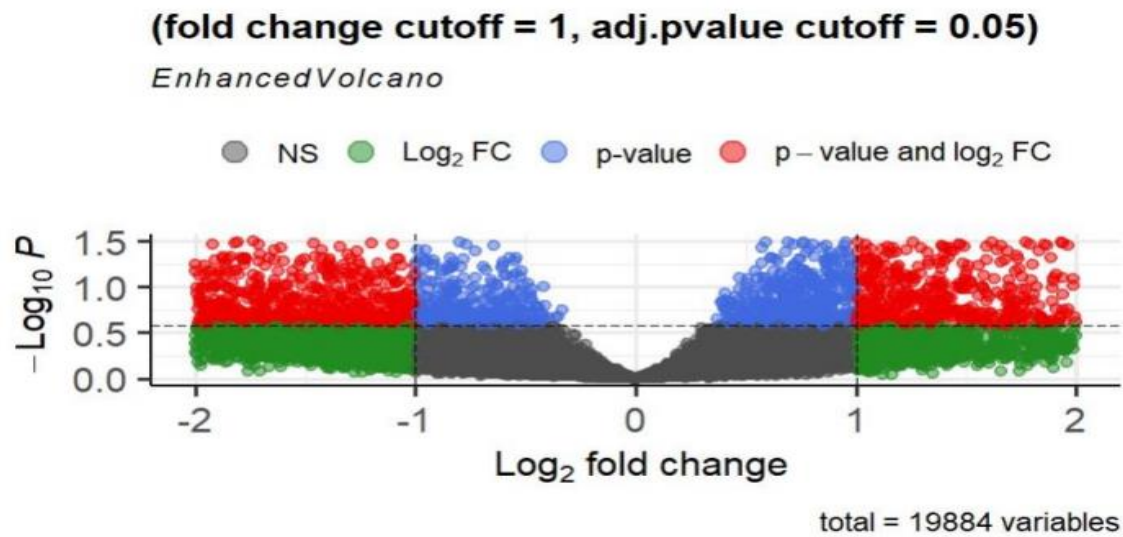


Figure 5 Volcano plot of significantly upregulated and downregulated genes identified by differential expression analysis

KEGG pathway enrichment analysis showed significant activation of immune and inflammation pathways such as cytokine-cytokine receptor interaction and Toll-like receptor signaling pathways. These data are consistent with a strong host immune response to infection with *Mycobacterium tuberculosis*.

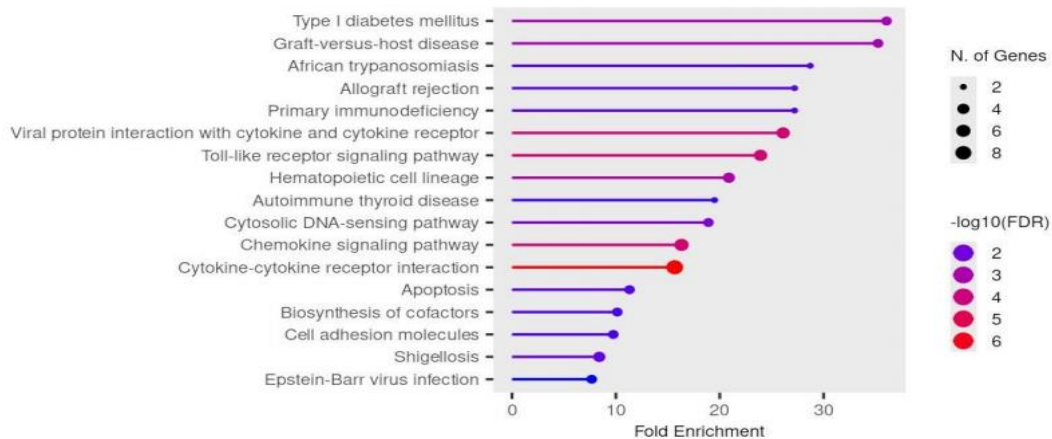


Figure 6. KEGG pathway enrichment analysis shows significantly enhanced immune and inflammatory pathways in spinal tuberculosis.

Enrichment map analysis showed strong interactions between immune-related pathways.

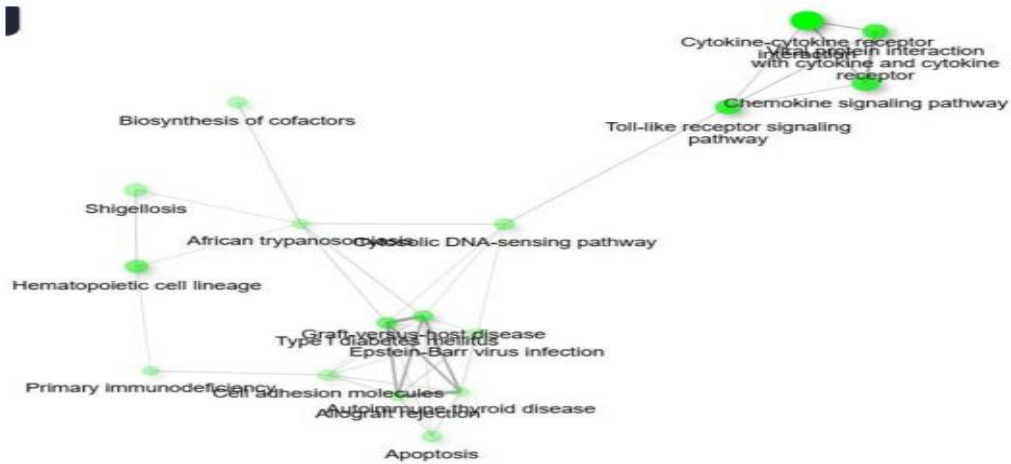


Figure 7. Enrichment map shows the interactions and overlaps between significantly enhanced biological pathways.

Protein-protein interaction analysis detected major hub proteins such as TNFRSF family proteins and the complement proteins such as C1QA, C1QB, C1QC, and FCN3.

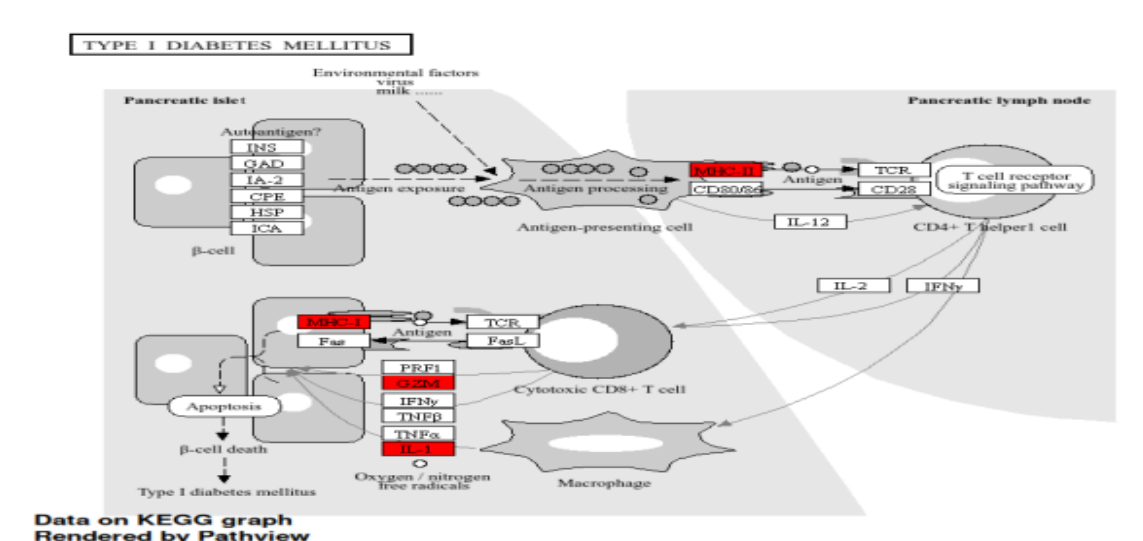


Figure 8. KEGG pathway map showing differentially expressed genes engaged in immune-related signaling pathways.

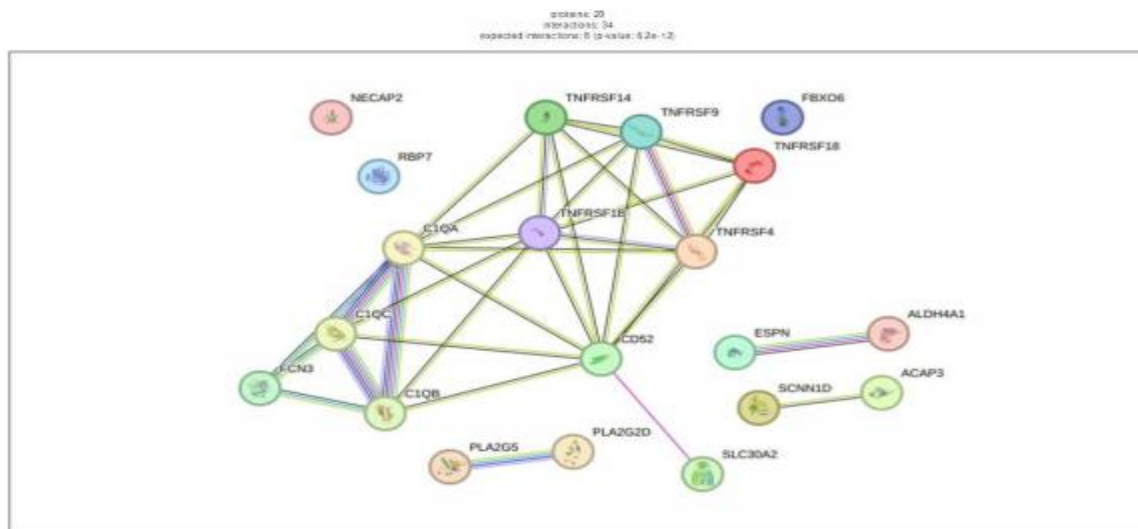


Figure 9. Protein-protein interaction (PPI) network of the top differentially expressed proteins related to spinal tuberculosis.

Conclusion

Finally, we explored transcriptomic alterations related to spinal tuberculosis using RNA-seq data from GSE268366. The discoveries revealed a strong connection between spinal

TB and problems with immune and inflammatory responses. We identified several important genes and pathways that could serve as potential biomarkers and treatment targets for future research.

References

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Appendix

Contribution table

Amira	Code execution and output of milestones 2,3 &4 alongside report writing of milestone 2
Inji	Report writing & reviewing of milestones 2 and 4
Sara & Tasabih	Report writing of milestone 3
Habiba & Menna	Report writing of all milestones and ppt generation